



Robust color image watermarking using invariant quaternion Legendre-Fourier moments

Khalid M. Hosny¹ · Mohamed M. Darwish²

Received: 1 August 2017 / Revised: 11 January 2018 / Accepted: 15 January 2018 /

Published online: 5 March 2018

© Springer Science+Business Media, LLC, part of Springer Nature 2018

Abstract In this paper, a geometrically invariant color image watermarking method using Quaternion Legendre-Fourier moments (QLFMs) is presented. A highly accurate, fast and numerically stable method is proposed to compute the QLFMs in polar coordinates. The proposed watermarking method consists of three main steps. First, the Arnold scrambling algorithm is applied to a binary watermark image. Second, the QLFMs of the original host color image are computed. Third, the binary digital watermark is embedding by performing the quantization of selected QLFMs. Two different groups of attacks are considered. The first group includes geometric attacks such as rotation, scaling and translation while the second group includes the common signal processing attacks such as image compression and noise. Experiments are performed where the performance of proposed method is compared with the existing moment-based watermarking methods. The proposed method is superior over all existing quaternion moment-based watermarking in terms of visual imperceptibility capability and robustness to different attacks.

Keywords Quaternion Legendre-Fourier moments · Color image watermarking · Geometric attacks · Rotation invariant

1 Introduction

Because of the rapid development advance of modern times, Internet and images, intellectual property and image copyright protection have become a prominent problem. Therefore, image watermarking technology has become a growing concern and information security has been extensively researched and used.

✉ Khalid M. Hosny
k_hosny@yahoo.com

¹ Department of Information Technology, Faculty of Computers and Informatics, Zagazig University, Zagazig 44519, Egypt

² Department of Mathematics, Faculty of Science, Assiut University, Assiut 71516, Egypt

Digital watermarks are used in copyright protection and securing data during their transmission through networked environment [7, 23]. Technically, digital watermarking aims to hide a small piece of digital data called a “digital watermark” into the actual digital media such as digital images, digital videos and digital audios without significance change of its normal usage [17, 47].

In order to ensure multimedia copyright protection, digital watermarking has been proposed as a modern technology used for this and consists in embedding an imperceptible information, yet detectable signal in digital multimedia content such as images or video, while it can be easily and correctly detected by the watermarking algorithm. Based on their specific security purposes, different methods of digital watermarking can be categorized into the following: robust, semi-fragile or fragile methods [6, 22, 24]. Digital image watermarking provides an efficient protection of digital information for various applications including telemedicine [27], e-health [4], medical images [25, 28, 48], online social networks [29], e-governance [15].

In the last decade, developing robust image watermarking algorithms have the great efforts by researchers [20]. Some of the researchers have been proposed watermarking scheme using green computing approach [30]. On the other hand, attacks on systems of image watermarking have become more complex and sophisticated. In general, these image watermarking systems attacks can be classified into geometric distortions and common image processing operations like noise [3].

The embedding, detection and extraction (retrieval) of digital watermark are the main steps in the watermarking process. The detection of the digital watermark usually encounters several problems such as geometric attacks. Due to its small size, any small attack may lead to failed detection of the embedded watermark. Therefore, robustness against geometric attacks is a very desirable characteristic.

During the last two decades, researchers paid their attention to watermarking technology based on invariants moments. Alghoniemy and Tewfik [1] first applied image moments to image watermarking technology which is robust against RST attacks by using the Hu’s moment invariants. Unfortunately, their method encountered major problems such as instability and poor visual imperceptibility. Since then, a several moment-based watermarking algorithms have been proposed.

Xin et al. [42] and Ismail et al. [14] used circular Zernike moments in image watermarking. Zhu et al. [50] utilized Radon transform and complex moments in RST invariant image watermarking. Wavelet moments and multi-scale Harris detector were proposed by Wang et al. [36] for image watermarking. Affine Legendre moment invariants were used by Zhang et al. [49] in image watermarking. Singh and Ranade [26] proposed a geometrically invariant and high capacity image watermarking scheme using accurate radial transform. Tsougenis et al. [32] discussed the flexibility of the image watermarking in polar harmonic transforms. Based on radial harmonic Fourier moments magnitude, Yang et al. [45] proposed a geometrically resilient digital watermarking scheme. Wang et al. [38] proposed a geometrically invariant image watermarking method based on radial harmonic Fourier moments (RHFMs). Recently, Hosny and Darwish [12] proposed an invariant image watermarking method using highly accurate polar harmonic transforms.

Image watermarking based on invariant moment is one of the most powerful (effective) schemes of watermarking. However, the existing watermarking schemes were mainly developed for grey-scale images.

Color images are more common and play an influential role in our everyday life. The color images provide us with more valuable information than grey-scale image. During the last

years, watermarking of color images received more attention from researchers around the world. The embedded digital watermark into color images may be a grey-scale or a color watermark. In [18, 34], the watermark is directly embedded into three color channels or the luminance component of color images. Therefore, the quality and robustness of watermark detection will be lowered because of losing some significant color information.

In [2], the pixel-wise masking watermarking technique was extended to color image, and the digital watermark was embedded in the Y component of the YCbCr model. Chou and Liu [5] proposed a perceptually optimized color image watermarking scheme by using wavelet coefficients of high perceptual redundancy to insert (embed) high-strength watermark signals. The watermarks are determined by inherent amount of perceptual redundancy in wavelet coefficients.

Liu et al. [16] proposed a wavelet-based watermarking scheme for color images through visual masking. A color visual model is designed to modify a perceptual model used in the image coding of grayscale images. Color image watermarking scheme based on non-blind luminance was presented by Hussein et al. [13]. Peng et al. [21] presented a support vector machines (SVMs) based image watermarking method for color images in multi-wavelet domain, in which the special frequency band and property of image in multi-wavelet domain are employed.

The color image watermarking schemes based on conventional methods ignored significant correlation between components of color images and then, the performance of the watermarking system was affected inevitably. To overcome this challenging problem, researchers started to design and develop models to deal with the color images holistically.

Quaternion algebra with orthogonal and non-orthogonal moments are merged where a set of quaternion moments are derived to represent color images by using the Red, Green and Blue (RGB) channels. Recently, quaternion- moment based watermarking methods of color images has been started as a promising area of research. Tsougenis et al. [33] introduced quaternion radial moments as the embedding domain of the color images for watermark embedding. Based on the orthogonal property quaternion exponent moments (QEMs), Wang et al. [37] studied the local QEMs as alternative moment domain for color image watermarking to improve robustness. In [39, 44], QEMs were used in the watermarking of color images.

Since the quaternion representation can deal with a color image in a holistic way, Niu et al. [19] proposed a color image watermarking scheme based on quaternion radial harmonic Fourier moments (QRHFM). Yang et al. [46] and Wang et al. [40] proposed robust color image watermarking schemes by using approximated quaternion polar harmonic transform (QPHTs).

In general, the embedded watermark of a robust image watermarking scheme must resist a wide variety of possible attacks. Attacks can be categorized as geometric distortions and common signal processing attacks. The set of common signal processing attacks include median filtering, contamination with different kind of additive noise, and JPEG compression. Rotation, scaling, and translation (RST) are geometric distortions; these attacks are more challenging than common signal processing attacks. During the recent years, many algorithms which have better performances against common signal processing attacks are proposed. However, dealing with geometric distortions is still an open problem.

Because of the changing size of color image or its orientations even slightly, the ability to receive and retrieve the watermark might be reduced dramatically. Some of existing schemes have weak resistance to mixed geometric transformation attacks and poor imperceptibility.

In this paper, we propose a geometrically invariant robust color image watermarking algorithm using quaternion Legendre Fourier moments (QLFM), which can be seen as the generalization of LFM for grey-scale images. Unlike the quaternion image moments, the kernels computation of LFM are accurately computed by using the recursion and exact

integration therefore, It has no numerical stability issue and are free from their inherent limitations. Moreover, the LFM coefficients are more suitable for robust image watermarking. The proposed technique is different from existing watermarking-based quaternion techniques, fast, highly accurate and numerically stable QLFM moments provided us a better visual imperceptibility and higher robustness against the geometric distortions and common signal processing attacks. Numerical experiments are conducted where the results of the proposed method show better performance than all other methods in terms of robustness to geometric attacks and visual imperceptibility.

The rest of this paper is organized as follows. Section 2 presents preliminaries about the quaternions and conventional LFM for grayscale images, and a brief description of the QLFM for color images. In Section 3, the rotation, scaling, and translation (RST) invariant property of QLFM is introduced. Section 4 discusses robust color image watermarking using QLFM. The simulation results in Section 5 show the performance of our scheme. Finally, Section 6 concludes this paper.

2 Preliminaries

2.1 Quaternion representation of a color image

A quaternion, q , was defined by Hamilton [9] as a generalization of the complex numbers:

$$q = a + bi + cj + dk \quad (1)$$

The real numbers, a, b, c , and d , are the components of q . The imaginary units, i, j , and k , are defined according to the following rules:

$$\begin{aligned} i^2 = j^2 = k^2 = ijk = -1, \\ ij = -ji = k, jk = -kj = i, ki = -ik = j \end{aligned} \quad (2)$$

A quaternion is called pure quaternion when, $a = 0$. Both conjugate and the modulus of a quaternion are defined as follows:

$$q^* = a - bi - cj - dk \quad (3)$$

$$|q| = \sqrt{a^2 + b^2 + c^2 + d^2} \quad (4)$$

Ell and Sangwine [8] used the quaternion to represent a color image, $f(x, y)$, as follows:

$$f(x, y) = f_R(x, y)i + f_G(x, y)j + f_B(x, y)k \quad (5)$$

Where $f_R(x, y)$, $f_G(x, y)$, and $f_B(x, y)$ represent the Red, Green and Blue components of the pixel respectively.

2.2 Orthogonal Legendre-Fourier moments

Recently, in their work, Xiao et al. [41] defined radial shifted Legendre moment (RSLMs) and their moment invariants for analysis and recognition of grey-scale images. The Legendre-

Fourier moments (LFMs) of order p and repetition q are orthogonal circular moments defined in polar coordinates as follows:

$$M_{pq} = \frac{(2p+1)}{\pi} \int_0^{2\pi} \int_0^1 f(r, \theta) [L_{pq}(r, \theta)]^* r dr d\theta, \quad (6)$$

The functions of the LFMs, $L_{pq}(r, \theta)$, are defined as follows:

$$L_{pq}(r, \theta) = \bar{P}_p(r) \exp(\hat{i}q\theta) \quad (7)$$

Where both number p and q are integers; the operator $[.]^*$ refers to the complex conjugate and $\hat{i} = \sqrt{-1}$. The real-valued substituted shifted Legendre polynomials $\bar{P}_p(r)$ are orthogonal in the interval $0 \leq r \leq 1$ and satisfy the following orthogonality relation [41]:

$$\int_0^1 \bar{P}_p(r) \bar{P}_t(r) r dr = \frac{1}{2(2p+1)} \delta_{pt} \quad (8)$$

The substituted shifted Legendre polynomials $\bar{P}_p(r)$ are defined as [41]:

$$\begin{aligned} \bar{P}_p(r) &= \sum_{k=0}^p (-1)^{p-k} \binom{p+k}{2k} \binom{2k}{k} r^{2k} \\ &= \sum_{k=0}^p (-1)^{p-k} \frac{(p+k)! r^{2k}}{(p-k)!(k!)^2} \end{aligned} \quad (9)$$

The real-valued substituted shifted Legendre polynomials $\bar{P}_p(r)$ obey the following recurrence relation:

$$\bar{P}_{p+1}(r) = \frac{2p+1}{p+1} (2r^2-1) \bar{P}_p(r) - \frac{p}{p+1} \bar{P}_{p-1}(r), \quad (10)$$

Where:

$$\bar{P}_0(r) = 1, \bar{P}_1(r) = 2r^2-1.$$

Based on orthogonality property, an image function $f(r, \theta)$ could be reconstructed using the LFMs as follows:

$$f(r, \theta) = \sum_{p=0}^{\infty} \sum_{q=-\infty}^{\infty} M_{pq} \bar{P}_p(r) e^{iq\theta} \quad (11)$$

Since the summation to infinity is impossible in computing environment, then, eq. (11) is rewritten as follows:

$$f(r, \theta) = \sum_{p=0}^{\text{Max}} \sum_{q=-\text{Max}}^{\text{Max}} M_{pq} \bar{P}_p(r) e^{iq\theta} \quad (12)$$

2.3 The quaternion Legendre Fourier moments

The right-side quaternion Legendre Fourier moments (QLFMs) of order p with repetition q defined as:

$$M_{pq}^R = \frac{2p+1}{\pi} \int_0^{2\pi} \int_0^1 f(r, \theta) \bar{P}_p(r) e^{-\mu q \theta} r dr d\theta \quad (13)$$

Where p and q are defined as ($p \leq p_{\max}, q \leq q_{\max}$). The unit pure quaternion μ is defined as $\mu = (i + j + k) / \sqrt{3}$. A color image is represented as follows:

$$f(r, \theta) = f_R(r, \theta)i + f_G(r, \theta)j + f_B(r, \theta)k \quad (14)$$

Based on the properties of quaternion algebra, and using eq. (15) in eq. (14), the right-side QLFMs could be represented using the RGB channels as follows:

$$M_{pq}^R = A_{pq}^R + iB_{pq}^R + jC_{pq}^R + kD_{pq}^R \quad (15)$$

Where:

$$\begin{aligned} A_{pq}^R &= -\frac{1}{\sqrt{3}} [\text{imag}(M_{pq}(f_R)) + \text{imag}(M_{pq}(f_G)) + \text{imag}(M_{pq}(f_B))] \\ B_{pq}^R &= \text{real}(M_{pq}(f_R)) + \frac{1}{\sqrt{3}} [\text{imag}(M_{pq}(f_B)) - \text{imag}(M_{pq}(f_G))] \\ C_{pq}^R &= \text{real}(M_{pq}(f_G)) + \frac{1}{\sqrt{3}} [\text{imag}(M_{pq}(f_R)) - \text{imag}(M_{pq}(f_B))] \\ D_{pq}^R &= \text{real}(M_{pq}(f_B)) + \frac{1}{\sqrt{3}} [\text{imag}(M_{pq}(f_G)) - \text{imag}(M_{pq}(f_R))] \end{aligned} \quad (16)$$

Where $M_{pq}(f_R)$, $M_{pq}(f_G)$, and $M_{pq}(f_B)$ represent LFM for the red-, green- and blue-channel respectively. Based on eq. (16), computing QLFMs is dependent on the computation of the conventional LFM for the three single-channel images. Consequently, exact computation of LFM results in exact QLFMs.

Original color image could be reconstructed by a finite number of QLFMs ($p \leq p_{\max}, q \leq q_{\max}$) using the following form:

$$f^{\text{reco.}}(r, \theta) = \hat{f}_A(r, \theta) + \hat{f}_B(r, \theta)i + \hat{f}_C(r, \theta)j + \hat{f}_D(r, \theta)k \quad (17)$$

Where:

$$\begin{aligned} \hat{f}_A(r, \theta) &= \text{real}(\hat{A}_{pq}^R) - \frac{1}{\sqrt{3}} [\text{imag}(\hat{B}_{pq}^R) + \text{imag}(\hat{C}_{pq}^R) + \text{imag}(\hat{D}_{pq}^R)] \\ \hat{f}_B(r, \theta) &= \text{real}(\hat{B}_{pq}^R) + \frac{1}{\sqrt{3}} [\text{imag}(\hat{A}_{pq}^R) + \text{imag}(\hat{C}_{pq}^R) - \text{imag}(\hat{D}_{pq}^R)] \\ \hat{f}_C(r, \theta) &= \text{real}(\hat{C}_{pq}^R) + \frac{1}{\sqrt{3}} [\text{imag}(\hat{A}_{pq}^R) - \text{imag}(\hat{B}_{pq}^R) + \text{imag}(\hat{D}_{pq}^R)] \\ \hat{f}_D(r, \theta) &= \text{real}(\hat{D}_{pq}^R) \\ &\quad + \frac{1}{\sqrt{3}} [\text{imag}(\hat{A}_{pq}^R) + \text{imag}(\hat{B}_{pq}^R) - \text{imag}(\hat{C}_{pq}^R)] \end{aligned} \quad (18)$$

The value of $\hat{f}_A(r, \theta)$ is very close to 0; and $\hat{f}_B(r, \theta)$, $\hat{f}_C(r, \theta)$ and $\hat{f}_D(r, \theta)$ represent the red, green and blue components of the reconstructed color image

respectively. The terms A_{pq}^R , B_{pq}^R , C_{pq}^R and D_{pq}^R represent the reconstruction matrix of A_{pq}^R , B_{pq}^R , C_{pq}^R and D_{pq}^R respectively.

$$\begin{aligned}\hat{A}_{pq}^R &= \sum_{p=0}^{\infty} \sum_{q=0}^{\infty} A_{pq}^R \bar{P}_p(r) e^{\mu q \theta} \approx \sum_{p=0}^{pmax} \sum_{q=0}^{qmax} A_{pq}^R \bar{P}_p(r) e^{\mu q \theta}, \\ \hat{B}_{pq}^R &= \sum_{p=0}^{\infty} \sum_{q=0}^{\infty} B_{pq}^R \bar{P}_p(r) e^{\mu q \theta} \approx \sum_{p=0}^{pmax} \sum_{q=0}^{qmax} B_{pq}^R \bar{P}_p(r) e^{\mu q \theta}, \\ \hat{C}_{pq}^R &= \sum_{p=0}^{\infty} \sum_{q=0}^{\infty} C_{pq}^R \bar{P}_p(r) e^{\mu q \theta} \approx \sum_{p=0}^{pmax} \sum_{q=0}^{qmax} C_{pq}^R \bar{P}_p(r) e^{\mu q \theta}, \\ \hat{D}_{pq}^R &= \sum_{p=0}^{\infty} \sum_{q=0}^{\infty} D_{pq}^R \bar{P}_p(r) e^{\mu q \theta} \approx \sum_{p=0}^{pmax} \sum_{q=0}^{qmax} D_{pq}^R \bar{P}_p(r) e^{\mu q \theta} \quad (19)\end{aligned}$$

Based on eq. (2), the right-side and the left-side QLFMs of order p with repetition q are not equals. The left-side QLFMs are defined as follows:

$$M_{pq}^L = \frac{2p+1}{\pi} \int_0^{2\pi} \int_0^1 \bar{P}_p(r) f(r, \theta) e^{-\mu q \theta} r dr d\theta \quad (20)$$

The left-side and right-side of QLFMs are related by the following form:

$$M_{pq}^L = -M_{p-q}^R \quad (21)$$

3 RST invariants of quaternion Legendre Fourier moments

In this subsection, we will derive and summarize the RST invariant property of QLFMs.

3.1 Rotation invariance

Invariance to the similarity transformations such as rotation and scaling, are very important in pattern recognition applications. However, QLFMs moment invariants are negatively affected by the approximation and geometrical errors. The QLFMs of the two images f^{rot} and f have the following relations:

$$M_{pq}^R(f^{\text{rot}}) = M_{pq}^R(f) e^{\mu q \alpha} \quad (22)$$

Where $M_{pq}^R(f^{\text{rot}})$ and $M_{pq}^R(f)$ are the QLFM of f^{rot} and f respectively. A similar relationship for the left-side QLFMs is written as follows:

$$M_{pq}^L(f^{\text{rot}}) = e^{\mu q \alpha} M_{pq}^L(f) \quad (23)$$

Since, $|e^{\mu q \alpha}| = 1$, then:

$$|M_{pq}^R(f^{\text{rot}})| = |M_{pq}^R(f^{\text{rot}}) e^{\mu q \alpha}| = |M_{pq}^R(f^{\text{rot}})| |e^{\mu q \alpha}| = |M_{pq}^R(f)|$$

The magnitude values of the QLFM moments are rotation invariants. For the proof of Rotation invariance, see Appendix 1.

3.2 Scaling invariance

Let f^S be the scaled version of the image f , we could express the scaling QLFMs using the following equation:

$$\begin{aligned} M_{pq}^R(f^S) &= \frac{2p+1}{\pi} \sum_{k=0}^p \left(\sum_{i=k}^p a^{2i+2} C_{pi} d_{ik} \int_0^{2\pi} \int_0^1 f(\hat{r}, \theta) \bar{P}_k(\hat{r}) e^{-\mu q \theta} \hat{r} d\hat{r} d\theta \right) \\ &= \sum_{k=0}^p \frac{2p+1}{2k+1} \left(\sum_{i=k}^p a^{2i+2} C_{pi} d_{ik} M_{pq}^R(f) \right) \end{aligned} \quad (24)$$

Where $M_{pq}^R(f^S)$ and $M_{pq}^R(f)$ are the QLFM of f^S and f respectively. For the proof, see Appendix 2. Use $p=0$ and $q=0$ in eq. (24), yields:

$$M_{00}^R(f^S) = a^2 M_{00}^R(f) \quad (25)$$

The scale invariants can be constructed as follows:

$$\varphi_{pq} = \sum_{k=0}^p \frac{2p+1}{2k+1} \left(\sum_{i=k}^p (M_{00}^R(f))^{-(i+1)} C_{pi} d_{ik} \right) M_{pq}^R(f) \quad (26)$$

3.3 Translation invariance

Translation invariance is achieved when the centroid of the color image is coinciding with the origin of coordinates. Suk and Flusser [31] defined the centroid (x_c, y_c) of RGB color images as follows:

$$\begin{aligned} x_c &= (m_{10}(f_R) + m_{10}(f_G) + m_{10}(f_B)) / m_{00}, \\ y_c &= (m_{01}(f_R) + m_{01}(f_G) + m_{01}(f_B)) / m_{00} \\ m_{00} &= m_{00}(f_R) + m_{00}(f_G) + m_{00}(f_B) \end{aligned} \quad (27)$$

Where $m_{00}(f_R)$, $m_{10}(f_R)$ and $m_{01}(f_R)$, are the first orders geometric moments of red channel. Similarly, $m_{00}(f_G)$, $m_{10}(f_G)$ and $m_{01}(f_G)$, and, $m_{00}(f_B)$, $m_{10}(f_B)$ and $m_{01}(f_B)$ are the first orders geometric moments of green and blue channels respectively. By

locating the coordinate origin at centroid(x_c, y_c), the central QLFM, which are invariant to translation, are defined as follows:

$$\bar{M}_{pq}^R = \frac{2p+1}{\pi} \int_0^{2\pi} \int_0^1 f(\bar{r}, \bar{\theta}) \bar{P}_p(r) e^{-\mu q \bar{r}} \bar{r} d\bar{r} d\bar{\theta} \quad (28)$$

Where $f(\bar{r}, \bar{\theta})$ is the image pixel which is corresponding to (x_c, y_c) .

4 Robust watermarking scheme

In this section, a robust geometrically invariant color watermarking method will be presented. The section divided into four subsections. All the relevant details of the proposed method for highly accurate computation of the QLFMs moments in polar coordinates by using exact integration of kernel are summarized the first subsection.

In the second subsection, embedding the watermark information into color host image by adaptively modulating the QLFMs coefficients magnitudes are presented. In the Extraction process, the retrieval of the digital watermark from the accurate invariant QLFMs coefficients magnitudes is discussed in the third subsection. The general diagram of the proposed color image watermarking framework is depicted in Fig. 1.

4.1 Exact computation of QLFMs in polar coordinates

Accurate and stable computation of the QLFMs moments is an essential step which results in accurate and robust watermarking algorithms. Since the QLFMs are defined in a circular domain, computational processes in polar coordinates are preferable. In this subsection, a summary of this accurate method will be presented.

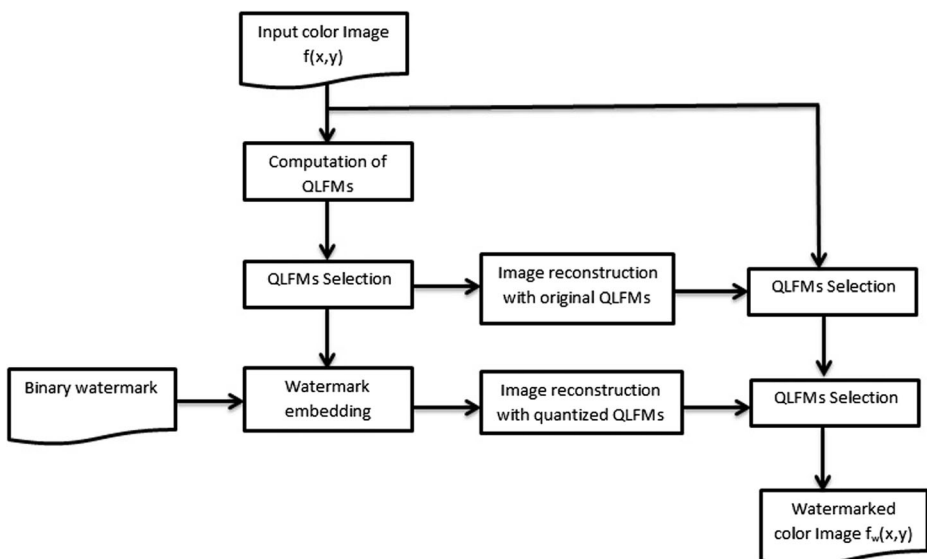


Fig. 1 Watermark embedding framework

For an input image of size $N \times N$ defined in Cartesian coordinates, the 2D image function $f(x, y)$ is converted onto the polar domain using a cubic interpolation [43]. All details of polar raster are found in [10, 11]. The exact LFM moments in polar coordinates are:

$$M_{pq} = \frac{2p+1}{\pi} \sum_i \sum_j \hat{f}(r_i, \theta_{ij}) L_{pq}(r_i, \theta_{ij}) \quad (29)$$

With:

$$L_{pq}(r_i, \theta_{ij}) = I_p(r_i) I_q(\theta_{ij}) \quad (30)$$

Where $\hat{f}(r_i, \theta_{ij})$ refers to the interpolated image intensity function and the kernels $I_p(r_i)$ and $I_q(\theta_{ij})$ are:

$$I_p(r_i) = \int_{U_i}^{U_{i+1}} \bar{P}_p(r) r dr = \frac{1}{4} \left[\frac{\bar{P}_{p+1}(r) - \bar{P}_{p-1}(r)}{(2p+1)} \right]_{U_i}^{U_{i+1}} \quad (31)$$

For $p \geq 1$. Based on eq. (10), we could write:

$$I_0(r_i) = \int_{U_i}^{U_{i+1}} (1) r dr = \frac{(U_{i+1}^2 - U_i^2)}{2}. \quad (32)$$

$$I_q(\theta_{ij}) = \int_{V_{ij}}^{V_{i,j+1}} e^{-i q \theta} d\theta \quad (33)$$

$$I_q(\theta_{i,j}) = \frac{\hat{i}}{q} \left(e^{-i q V_{i,j+1}} - e^{-i q V_{i,j}} \right) \quad (34)$$

For $q \geq 1$:

$$I_0(\theta_{i,j}) = (V_{i,j+1} - V_{i,j}) \quad (35)$$

Where:

$$U_{i+1} = R_i + \Delta R_i / 2; \quad U_i = R_i - \Delta R_i / 2; \quad (36)$$

$$V_{i,j+1} = \theta_{i,j} + \Delta \theta_{i,j} / 2; \quad V_{i,j} = \theta_{i,j} - \Delta \theta_{i,j} / 2 \quad (37)$$

Where R_i refers to the radial distance of circular ring.

Exact computation of QLFMs required the exact computation of LFM moments for each individual RGB component, $\hat{f}_R(r_i, \theta_{ij})$, $\hat{f}_G(r_i, \theta_{ij})$, and $\hat{f}_B(r_i, \theta_{ij})$ of the interpolated color image, $\hat{f}(r_i, \theta_{ij})$, then use these moments, $M_{pq}(\hat{f}_R(r_i, \theta_{ij}))$, $M_{pq}(\hat{f}_G(r_i, \theta_{ij}))$, and $M_{pq}(\hat{f}_B(r_i, \theta_{ij}))$ in eqs. (16) and (17).

4.2 Watermark embedding

The host color image function is $H = \{h(x, y), 0 < x \leq N, 0 < y \leq M\}$ where $h(x, y)$ represents the pixel value at the position (x, y) . Let $B = \{b(i, j) \in \{0, 1\}, 0 \leq i < P, 0 \leq j < Q\}$ be the binary watermark image to be inserted/embedded into the original host color image. The steps of embedding the digital watermark are discussed in the following subsections.

4.2.1 Watermark transforming

In order to improve the robustness of the digital watermark against the common different attacks, the pixel space relationship of the binary watermark image was dispelled and the binary watermark image was scrambled from B to B_1 by using a watermark scrambling algorithm Arnold transform [46].

$$B_1 = \{b_1(i, j) \in \{0, 1\}, 0 \leq i < P, 0 \leq j < Q\}.$$

As a result, it was transformed into a one-dimensional bit sequence as follows.

$$B_2 = \{b_2(l) = b_1(i, j) \in \{0, 1\}, 0 \leq i < P, 0 \leq j < Q, l = i \times Q + j\}.$$

4.2.2 Selection of QLFMs moments:

The accurate QLFMs moments of the original host color image F were computed by using the method described in section 4.1. The robustness of the proposed watermarking algorithm is enhanced by selecting the most suitable QLFMs moments based on two factors:

- The first one, QLFMs moments with $q = 4m, m \in \mathbb{Z}$ (i.e. $q = 0, q = 4, q = 8, q = 12, \dots$) are dropped from the selection process where are not suitable for encoding watermark bits [42].
- The second factor, only the independent QLFMs moments with positive repetition $q > 0$ are used. The QLFMs moments with negative repetition $q < 0$ are dependent, and then are dropped to avoid information redundancy.

Therefore, according to geometric invariance and the reconstruction accuracy of QLFMs coefficients, the independent and accurate final moment set used for watermark embedding in the proposed scheme based on selection process could be described as follows:

$$S = \{M_{pq}, q \neq 4m, m \in \mathbb{Z}\}$$

For a watermark bit sequence of length equal to $l = P \times Q$, the performance of the watermarking algorithm could be increased by selecting from the feature vector:

$$M(l) = \{M_{p_1 q_1}, M_{p_2 q_2}, \dots, M_{p_l q_l}\}$$

4.2.3 Embedding of digital watermark:

After the selection of the moments; the binary watermark could be embedded by modifying the magnitudes of these moments. For the sequence B_2 , the moments, $M(l)$, are selected from the

moments set S then the bits of the digital watermark are embedded by modifying the magnitude of the moments by using the dither modulation function [42]:

$$M'(l) = \left[\frac{M(l) - d_k(b2(l))}{\Delta} \right] * \Delta + d_k(b2(l)) \quad (38)$$

$$0 \leq l < P \times Q, \quad d_k(1) = \frac{\Delta}{2} + d_k(0), \quad d_k(0) \in [0, 1]$$

Where $M(l)$ are the unmodified QLFMs moments of host image while $M'(l)$ are the modified QLFMs moments' and the quantized version of $M(l)$. The $[\cdot]$ and Δ and $d_k(\cdot)$ refer to the rounding operator, the quantization steps and the dither function associated with the K key, respectively.

4.2.4 Reconstruction of the watermarked color image:

The watermarked color image consisted of two components, the first one was represented by the image formed from unchanged QLFMs' moments and defined by using the following form [46]:

$$f_{\text{rem}}(r, \theta) = f(r, \theta) - f_M(r, \theta) \quad (39)$$

With:

$$f_M(r, \theta) = \sum_{i=0}^{P \times Q - 1} M_{p_i q_i} L_{p_i q_i}(r, \theta) + M_{p_i, -q_i} L_{p_i, -q_i}(r, \theta) \quad (40)$$

While the second one is formed by the selected modified QLFMs' moments:

$$f_{M'}(r, \theta) = \sum_{i=0}^{P \times Q - 1} M'_{p_i q_i} L_{p_i q_i}(r, \theta) + M'_{p_i, -q_i} L_{p_i, -q_i}(r, \theta) \quad (41)$$

As a result, the watermarking image $f_w(r, \theta)$ could be obtained by embedding the compensation image on the host color image where $f_w(r, \theta)$ was formed as:

$$f_w(r, \theta) = f_{\text{rem}}(r, \theta) + f_{M'}(r, \theta) \quad (42)$$

4.3 Watermark extraction

Watermark extraction scheme is used to extract the binary watermark information from the selected coefficients of f_w . The extraction of watermark procedure in the proposed scheme does not need the original image where the watermark extraction process at the detector's side consists of almost the same processing steps with those of the embedding stage.

4.3.1 Selection of accurate feature vector

The QLFMs moments of the possibly attacked watermarked color image f_w were computed by using the proposed method (see Sec. 4.1). The selection of QLFMs moments in extraction and embedding processes were similar. Thus, the QLFMs feature vector, $M^*(l) = \{M^*_{p_1 q_1}, M^*_{p_2 q_2}, \dots, M^*_{p_l q_l}\}$, of a watermarked image f_w

can be selected by the same key K as in the watermark embedding procedure that carry the watermark information.

4.3.2 Binary watermark extraction:

Using the same Δ and quantizer as in the embedding process equation (38), we quantize the magnitude of each M'_i with the two dithers respectively:

$$|M'(l)|_j = \left\lceil \frac{|M^*(l)| - d_k(j)}{\Delta} \right\rceil * \Delta + d_k(j), \quad j = 0, 1 \quad (43)$$

By comparing the distances between $M'(l)$ and its quantized version $|M^*(l)|$, we receive:

$$\hat{b}_2(l) = \underset{j \in [0,1]}{\operatorname{argmin}} \left(|M'(l)|_j - |M^*(l)| \right)^2 \quad (44)$$

Where $|M'(l)|_j$ is the i th QLFMs of the attacked color image which is quantized considering a bit value of $j \in [0, 1]$. Therefore, a bit $\hat{b}_2(l)$ is decided to be 0 or 1 according to the distance between the corresponding quantized QLFMs and its original value $M^*(l)$. The extracted bit is assigned depending on the j of the minimum distance $\hat{b}_2(l)$ value. The one-dimensional binary sequence was extracted from QLFMs magnitudes as follows [46]:

Where $\hat{B}_2 = \{ \hat{b}_2(l) \in \{0, 1\}, 0 \leq l < P \times Q \}$ was transformed to represent the binary watermark image. As a result, the binary image was descrambled by using inverse Arnold transform to form the binary watermark image.

$$\hat{B}_1 = \{ \hat{b}_1(i, j) \in \{0, 1\}, 0 \leq i < P, 0 \leq j < Q \}$$

5 Numerical experiments

In order to analyse and evaluate the performance of the proposed method and compare it with the existing color watermarking algorithms in term of visual imperceptibility and watermark robustness. The watermark invisibility is measured by peak signal-to-noise ratio (PSNR) and the structural similarity image index (SSIM), while watermark robustness is measured as the bit error rate (BER) of extracted watermark. A set of numerical experiments was performed. In the first subsection, experiments for watermark invisibility of different host color images are described to show the transparency of the proposed method. Robustness against various attacks discussed in the second subsection. Average elapsed CPU times for the proposed method and other existing watermarking schemes are presented in the third subsection.

In the conducted experiments, eight real color images of size 256×256 as displayed in Fig. 2 were used as host test images and two kind of test watermarks, first one each of dimensions 32×32 , are shown in Figs. (3.a)-(3.b) and the second watermark images of size 10×10 are displayed in Figs. (3.c)-(3.l).

The embedding strength in all the conducted numerical experiments was $\Delta = 0.3$, maximum order of moment was 40 and the watermark information was a binary sequence of 128–256-



Fig. 2 Color images used in Testing the watermarking methods

512 bits. Also, the obtained results were compared with the results of the recent schemes QEMs [39], QRHFM [19] and QPHTs [46].

5.1 Watermark invisibility

The invisibility of the embedded watermark is one of the important criteria to evaluate the performance of watermarking algorithm. The peak signal-to-noise ratio (PSNR) usually used as a quantitative measure for the watermarked image, f_w . The PSNR defined as follows:

$$\text{PSNR}(f, f_w) = 10 \log_{10} \frac{255^2}{\text{MSE}} \quad (45)$$

Where MSE is the mean square error and defined as:

$$\text{MSE} = \frac{1}{N^2} \left(\sum_{i=1}^N \sum_{j=1}^N [f_w(i, j) - f(i, j)]^2 \right) \quad (46)$$

Figures 4 and 5 shows the result of applying the proposed method for 128-bit watermark sequence embedding in sample color images with size 256×256 pixels with quantization step Δ varying from 0.1 to 1.0. In comparison with other quaternion moment-based watermarking algorithms such as QEMs [39], QRHFM [19] and QPHTs [46], it is observed from Fig. 4 that, the average value of PSNR decreased as the quantization step Δ increased.

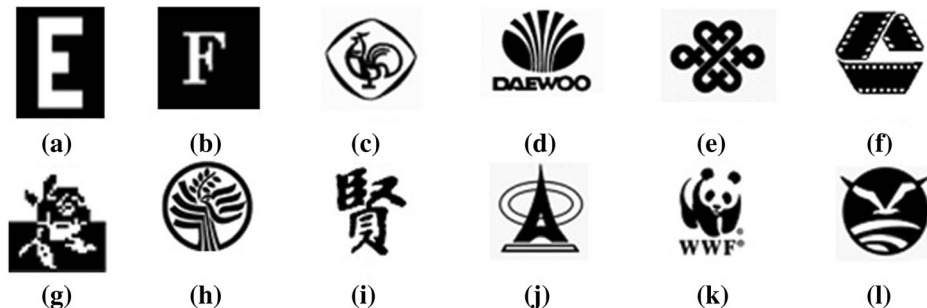


Fig. 3 Binary watermark images

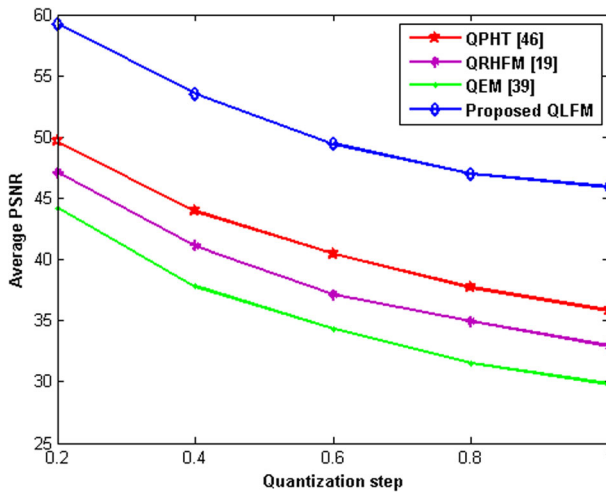


Fig. 4 Visual imperceptibility relative to quantization step Δ

Figure 5 shows the watermarked images with their corresponding PSNR values after embedding 128-bit watermarks information into Lena color image with size 256X256 pixels by using quaternion-based watermarking [19, 39, 46] and the proposed approaches with quantization step $\Delta = 0.3$.

It can be seen clearly that the proposed algorithm has a better performance in terms of visual imperceptibility and the subjective quality of watermarked color image than compared algorithms.

In order to evaluate the watermarked image quantitatively, an additional experiment is performed where the SSIM [35] is used to measure quality of the watermarked image. The SSIM is defined as follow:

$$\text{SSIM}(f, f_w) = \frac{(2\mu_f\mu_{f_w} + C_1)(2\sigma_{ff_w} + C_2)}{(\mu_f^2 + \mu_{f_w}^2 + C_1)(\sigma_f^2 + \sigma_{f_w}^2 + C_2)} \quad (47)$$

Where μ_f and μ_{f_w} are the average luminance values of original image f and the watermarked image f_w , respectively; σ_f and σ_{f_w} are the standard variance of f and f_w ,



Fig. 5 Reconstructed watermarked color image of Lena using the proposed method and the existing methods [19, 39, 46]

Table 1 The average PSNR and SSIM values with Different values of quantization step (Δ)

Q.S.	The proposed Method		QRHFM [19]		QPHTs [46]		QEMs [39]	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
0.2	59.24	0.9999	47.09	0.998	49.66	0.998	44.20	0.9922
0.4	53.51	0.9997	41.10	0.9918	43.91	0.9918	37.78	0.9707
0.6	49.44	0.9993	37.10	0.9811	40.46	0.9811	34.31	0.9474
0.8	46.99	0.9987	34.96	0.9692	37.69	0.9692	31.51	0.899
1.0	45.88	0.9984	32.91	0.9594	35.82	0.9594	29.81	0.8686

respectively; σ_{ff_w} is the covariance between f and f_w ; C_1 and C_2 are small fixed positive constants and adopted to avoid the denominators from being zero.

Table 1 represents the average values of PSNR and average values of SSIM of the proposed and different watermarking schemes with selected value of watermark embedding strength length {quantization step (Δ)}. As can be seen from Table 1, the proposed scheme is better

Attacks		Proposed QLFM	QEMs [39]	QRHFM [19]	QPHTs [46]
Rotation angle	5				
		BER=0.0273	BER=0.2051	BER=0.1465	BER=0.0742
	15				
		BER=0.0293	BER=0.1660	BER=0.1016	BER=0.0742
	25				
		BER=0.0283	BER=0.178	BER=0.091	BER=0.050
	45				
		BER=0.0273	BER=0.1738	BER=0.1152	BER=0.0615
Scaling factor	0.75				
		BER=0.0225	BER=0.1846	BER=0.1123	BER=0.0732
	1.25				
		BER=0.0068	BER=0.1602	BER=0.0498	BER=0.0625
	1.5				
		BER=0.0078	BER=0.1357	BER=0.0693	BER=0.0488
	2				
		BER=0.0088	BER=0.1309	BER=0.0566	BER=0.0469
Translation (H 5, V 15)					
		BER=0.0029	BER=0.0361	BER=0.0293	BER=0.0107

Fig. 6 The extracted binary watermarks under geometric attacks

than their corresponding schemes by QEMs [39], QRHFM [19] and QPHTs [40, 46] in terms of visual imperceptibility and the transparency.

5.2 Watermark robustness

Watermark robustness of any watermarking algorithm is measured quantitatively by using bit error rate (BER) of extracted watermark which is defined as:

$$\text{BER} = \frac{B_{\text{error}}}{l} \quad (48)$$

Where $l = P \times Q$ represents the total number of embedded bits or the dimensions of the watermark image while B_{error} refers to the number of incorrectly extracted bits. The robustness of the proposed watermarking algorithm is tested and evaluated in terms of geometric attacks (rotation, scaling, translation and shearing) and common image processing attacks (JPEG compression, median and Gaussian filtering, different type of noise).

Embedding capacity bits are 128, 256 and 512. Extensive numerical experiments showed that, 128 represents the best choice of embedding capacity [12]. Consequently, the 128-bit embedding capacity is used in the conducted numerical experiments. It is observed that the proposed algorithm has higher robustness against various attacks rotation, scaling and JPEG compression with different embedding capacity.

Additional simulation results, which are obtained by the proposed watermarking scheme, for geometric distortions and common image processing operations are depicted in Figs. 6 and 7.

Figure 6 shows the extracted watermark images and their corresponding BER values using the proposed QLFMs watermarking algorithm and the other quaternion moment-based watermarking methods [19, 39, 46] for standard color image “Lena” and the standard binary image of Letter “E” with size 32×32 . It is easy to notice that the

Attacks		Proposed QLFM	QEMs [39]	QRHFM's [19]	QPHT's [46]
JPEG Compression	70				
		BER=0.0195	BER= 0.1045	BER=0.0391	BER=0.1035
	80				
		BER=0.0098	BER= 0.1201	BER= 0.0273	BER=0.0898
	90				
		BER=0.0039	BER= 0.1270	BER= 0.0313	BER=0.0605
Salt & Peppers Noise (0.01)					
		BER=0.0127	BER=0.0957	BER=0.0830	BER=0.0703
Gaussian noise (0.01)					
		BER=0.0117	BER=0.0703	BER=0.0635	BER=0.0566

Fig. 7 Watermark robustness to common image processing attacks

extracted watermark images are very similar to the original binary watermark image as displayed in Fig. 3a. It can be seen clearly that the proposed scheme has higher robustness to geometric distortions.

In Fig. 7, the extracted watermark images “E” and its BER values are shown, which are obtained by the proposed QLFMs scheme and the other quaternion moment-based watermarking methods [19, 39, 46] to evaluate the robustness of the proposed watermarking to common image processing. In general, the results of the performed numerical experiments show the robustness of the proposed QLFMs-based watermarking algorithms against different kinds of attacks and ensure its robustness over most common attacks.

Table 2 shows the average BER values are computed accurately after embedding and extracting a 128-bit watermark using the proposed QLFMs and those obtained with QPHTs [46], QRHFM [19], and QEMs [39] methods against different common various attacks. As shown in the table, the proposed watermarking algorithm is robust against most of the attacks in terms of BER.

In the next numerical experiment, the test watermark as displayed in Fig. 3a are embedded in the host color images as displayed in Fig. 2 by using the proposed watermarking scheme and the other quaternion moment-based watermarking methods [19, 39, 46]. Then, the watermarked color images are generated. For each watermarked color image, each watermark is extracted under geometric distortions and different common attacks (rotation, scaling, translation, median filtering of size 3×3 , and different kinds of noise namely, Gaussian noise, Salt & Peppers noise). The results were obtained where the average values of BERs of the retrieved watermarks from each watermarked color images are summarized and listed in Table 3. These results clearly show that proposed method is invariant to all of these image

Table 2 Average BER values of the proposed method compared with the existing methods [19,39,46] for various attacks

Attacks		Proposed	QPHT [46]	QRHFM [19]	QEM [39]
Rotation angle	5°	0	0.0052	0.0060	0.0104
	15°	0	0.0112	0.0095	0.0104
	25°	0	0.0086	0.0138	0.0164
	35°	0	0.0069	0.0078	0.0095
	45°	0	0.0095	0.0156	0.0208
Scaling factor	0.25	0.2981	0.4383	0.3020	0.4288
	0.5	0.0234	0.0546	0.0434	0.0885
	0.75	0.0039	0.0112	0.0049	0.0234
	1.25	0	0.0078	0.0060	0.0069
	1.5	0	0.0112	0.0060	0.0121
	1.75	0	0.0078	0.0060	0.0130
	2.0	0	0.0112	0.0130	0.0138
JPEG Compression ratio	0	0.4380	0.5060	0.4557	0.4913
	10	0.3326	0.3837	0.3406	0.3915
	20	0.0625	0.1753	0.1021	0.1267
	30	0.0234	0.0824	0.0369	0.0573
	40	0.0104	0.0312	0.0122	0.0173
	50	0.0049	0.0121	0.0086	0.0117
	60	0	0.0069	0.0086	0.0069
	70	0	0.0069	0.0086	0.0034
	80	0	0.0138	0.0086	0.0078
	90	0	0.0078	0.0112	0.0069
Shearing (0%-1%)		0	0.0107	0.0205	0.0098

processing attacks. Its performance is better than other aforementioned moment-based watermarking algorithms.

Table 3 clearly shows that, the proposed watermarking algorithm is invariant to all geometric distortion attacks and robust against image processing attacks. On the other side, QPHTs [46], QRHFM [19], and QEMs [39], and show little robustness against scaling attacks.

To sum up, the above experimental results proved the validity of our scheme and its higher robustness against geometric transformations and image processing attacks compared to alternative color watermarking methods in the literature. This is because that the highly accurate and stable QLFMs and the QLFMs magnitudes are invariant to geometric transformation, so QLFMs are more suitable for robust color image watermarking.

5.3 Computational CPU times

Computational CPU times are essential issues in evaluating new computational methods. Low computational CPU times are used as indicators that reflected the efficiency of the proposed algorithm. The computational CPU times required by the proposed watermarking algorithm are estimated and compared with the other computational CPU times required by the existing moment-based watermarking algorithms in a quantitative fashion. In order to ensure the efficiency of the proposed algorithm, numerical experiments are performed with different color images as displayed in Fig. 2. In these experiments, different moments are computed for moment order ranging from 0 to 30 with increment value of 10 by using the proposed method and other existing methods [19, 39, 46]. The average elapsed CPU times for these methods are

Table 3 Average BER values of the watermark detection for common attacks

Attacks		Proposed	QPHT [46]	QRHFM [19]	QEM [39]
Rotation angle	5°	0	0	0.0218	0
	15°	0	0	0.0211	0
	45°	0	0	0.0221	0
	70°	0	0	0.0215	0.0074
	90°	0	0	0.0052	0.0029
Scaling factor	0.5	0.01	0.0130	0.0303	0.0313
	0.75	0	0.0102	0.0179	0.0104
	1.25	0	0.0182	0.0117	0.0125
	1.5	0	0.0039	0.0280	0.0039
	2.0	0	0	0.0277	0
Translation	(H 2,V 15)	0	0	0.0192	0.0117
	(H 20,V 20)	0	0	0.0192	0.0309
	(H 15,V 2)	0	0	0.0192	0.0117
	(H 50,V 0)	0	0	0.0192	0.0128
	(H 0,V 50)	0	0	0.0192	0.0128
JPEG Compression ration	30	0.012	0.026	0.1102	0.0428
	40	0.0098	0.0117	0.0835	0.0313
	50	0	0.0078	0.0576	0.0195
	70	0	0.0052	0.0416	0.0026
	90	0	0	0.0576	0.0039
Salt and Peppers Noise (0.01)		0	0.0026	0.0351	0
Gaussian Noise (0.01)		0.0013	0.0091	0.0536	0.0173
Median Filtering (3 × 3)		0.0020	0.0117	0.0566	0.0078

Table 4 Average CPU times in seconds for proposed and the existing methods

Moment Order	The Proposed method	QPHTs [46]	QRHFM [19]	QEMs [39]
0	0.0130	0.0482	0.1205	0.0521
10	0.5580	9.2031	12.6045	10.4023
20	2.1923	24.9852	42.0231	27.0984
30	5.0432	52.9446	67.0241	54.2045

shown in Table 4. It clearly shows that the proposed method is faster than other moment-based watermarking methods.

In order to interpret these results, the watermarking process consists of few main steps such as moment computation for host images, quantization, embedding, reconstruction of the host image, and watermark extraction. Unlike all existing moment-based watermarking methods, the proposed method employed an accurate kernel-based method where these kernels are image independents where the elapsed CPU times required for computing moments are much smaller than the corresponding times required by the existing methods.

6 Conclusion

The Design of robust color image watermarking techniques is still challenging especially against common geometric distortions. In this paper, a new robust geometrically invariant color watermarking algorithm based on fast and accurate QLFMs is proposed, which scrambled binary watermarks by using scrambling transform called Arnold scrambling transform to give a reliable performance in terms of robustness and visual imperceptibility. Based on the distribution of moment magnitude, the more accurate and robust QLFMs were computed and selected. Detailed analysis and discussions have been presented to compute the accurate QLFMs. Then, the binary watermark is embedded by modifying their invariant QLFMs magnitudes using dither modulation function. Based on the invariants properties of QLFMs and quantization process, our proposed method is very robust to variety attacks.

The results of numerical experiments obviously show higher robustness of the proposed watermarking algorithms against different geometric distortions such as rotation and scaling and image processing attacks such as JPEG compression, median filtering, and different types of noise. The performance of the proposed algorithm outperformed existing quaternion moment-based watermarking schemes in terms of visual imperceptibility and robustness.

Appendix 1

A counter-clockwise rotation with an angle α , the transformed image intensity function is defined as:

$$f^{\text{rot}}(r, \theta) = f(r, \theta + \alpha) \quad (49)$$

Assume $\hat{\theta} = \theta + \alpha$, then $\theta = \hat{\theta} - \alpha$, $d\theta = d\hat{\theta}$. The QLFMs of the two images f^{rot} and f have the following relations:

$$\begin{aligned}
M_{pq}^R(f^{\text{rot}}) &= \frac{2p+1}{\pi} \int_0^{2\pi} \int_0^1 f^{\text{rot}}(r, \theta) [L_{pq}(r, \theta)]^* r dr d\theta \\
&= \frac{2p+1}{\pi} \int_0^{2\pi} \int_0^1 f(r, \theta + \alpha) \bar{P}_p(r) e^{-\mu q \theta} r dr d\theta \\
&= \frac{2p+1}{\pi} \int_0^{2\pi} \int_0^1 f(r, \hat{\theta}) \bar{P}_p(r) e^{-\mu q \left(\hat{\theta} - \alpha\right)} r dr d\hat{\theta} \\
&= \frac{2p+1}{\pi} \int_0^{2\pi} \int_0^1 f(r, \hat{\theta}) \bar{P}_p(r) e^{-\mu q \hat{\theta}} e^{\mu q \alpha} r dr d\hat{\theta} \\
&= M_{pq}^R(f) e^{\mu q \alpha}
\end{aligned} \tag{50}$$

Where $M_{pq}^R(f^{\text{rot}})$ and $M_{pq}^R(f)$ are the QLFM of f^{rot} and f respectively.

Appendix 2

Let f^S be the scaled version of the image f . Let $f^S(r, \theta) = f\left(\frac{r}{a}, \theta\right)$ be the color image expanded by the scale factor a , and $\hat{r} = \frac{r}{a}$, then $dr = a d\hat{r}$, we could express the scaling QLFMs using the following equation:

$$\begin{aligned}
M_{pq}^R(f^S) &= \frac{2p+1}{\pi} \int_0^{2\pi} \int_0^1 f^S(r, \theta) [L_{pq}(r, \theta)]^* r dr d\theta \\
&= \frac{2p+1}{\pi} \int_0^{2\pi} \int_0^1 f^S(r, \theta) \bar{P}_p(r) e^{-\mu q \theta} r dr d\theta \\
&= \frac{2p+1}{\pi} \int_0^{2\pi} \int_0^1 f\left(\frac{r}{a}, \theta\right) \bar{P}_p(r) e^{-\mu q \theta} r dr d\theta \\
&= \frac{2p+1}{\pi} \int_0^{2\pi} \int_0^1 f(\hat{r}, \theta) \bar{P}_p(a\hat{r}) e^{-\mu q \theta} a^2 \hat{r} d\hat{r} d\theta \\
&= a^2 \frac{2p+1}{\pi} \int_0^{2\pi} \int_0^1 f(\hat{r}, \theta) \bar{P}_p(a\hat{r}) e^{-\mu q \theta} \hat{r} d\hat{r} d\theta
\end{aligned} \tag{51}$$

Where $\bar{P}_p(a\hat{r})$ is scaled version of $\bar{P}_p(\hat{r})$. The scaled substituted shifted Legendre polynomials $\bar{P}_p(a\hat{r})$ could be expressed in terms of the substituted shifted Legendre polynomials $\bar{P}_p(\hat{r})$ as follows:

$$\bar{P}_p(a\hat{r}) = \sum_{k=0}^p \left(\sum_{i=k}^p a^{2i} C_{pi} d_{ik} \right) \bar{P}_k(\hat{r}) \tag{52}$$

Where:

$$C_{pi} = (-1)^{p-k} \frac{(p+k)!}{(p-k)!(k!)^2} \tag{53}$$

$$d_{pi} = \frac{(2k+1)(k!)^2}{(p+k+1)!(p-k)!} \tag{54}$$

Substitution equation (52) into equation (51) yields:

$$\begin{aligned} M_{pq}^R(f^S) &= \frac{2p+1}{\pi} \sum_{k=0}^p \left(\sum_{i=k}^p a^{2i+2} C_{pi} d_{ik} \int_0^{2\pi} \int_0^1 f(\hat{r}, \theta) \bar{P}_k(\hat{r}) e^{-i\mu q \theta} \hat{r} d\hat{r} d\theta \right) \\ &= \sum_{k=0}^p \frac{2p+1}{2k+1} \left(\sum_{i=k}^p a^{2i+2} C_{pi} d_{ik} M_{pq}^R(f) \right) \end{aligned} \quad (55)$$

Where $M_{pq}^R(f^S)$ and $M_{pq}^R(f)$ are the QLFM of f^S and f respectively. Use $p=0$ and $q=0$ in equation (55), yields:

$$M_{00}^R(f^S) = a^2 M_{00}^R(f) \quad (56)$$

The scale invariants can be constructed as follows:

$$\varphi_{pq} = \sum_{k=0}^p \frac{2p+1}{2k+1} \left(\sum_{i=k}^p (M_{00}^R(f))^{-(i+1)} C_{pi} d_{ik} \right) M_{pq}^R(f) \quad (57)$$

References

- Alghoniemy M, Tewfik AH (2004) Geometric invariance in image watermarking. *IEEE Trans Image Process* 13(2):145–153
- Al-Otumi HA, Al-Taba AO (2009) Adaptive color image watermarking based on a modified improved pixel-wise masking technique. *Comput Electr Eng* 35(5):673–695
- Bianchi T (2013) Secure watermarking for multimedia content protection: a review of its benefits and open issues. *IEEE Signal Process Mag* 30(2):87–96
- Chauhan DS, Singh AK, Kumar B, Saini JP (2017) Quantization based multiple medical information watermarking for secure e-health. *Multimedia Tools and Applications*:1–13. <https://doi.org/10.1007/s11042-017-4886-4>
- Chou CH, Liu KC (2010) A perceptually tuned watermarking scheme for color images. *IEEE Trans Image Process* 19(11):2966–2982
- Chu SC, Huang HC, Shi Y, Wu SY, Shieh CS (2008) Genetic watermarking for Zerotree-based applications. *Circuits, Syst, Signal Process* 27(2):171–182
- Cox IJ, Millter ML, Bloom JA, Fridrich J, and Kalker T (2008) *Digital watermarking and steganography*. Morgan Kaufmann Publishers (Elsevier), Burlington,
- Ell TA, Sangwine SJ (2007) Hypercomplex Fourier transforms of color images. *IEEE Trans Image Process* 16:22–35
- Hamilton WR (1866) *Elements of quaternions*. Longmans Green, London
- Hosny KM (2011) Accurate orthogonal circular moment invariants of gray-level images. *J Comput Sci* 7(5):715–722
- Hosny KM, Darwish MM (2016) A Kernel-Based method for Fast and accurate computation of PHT in polar coordinates, *Journal of Real-Time Image Process.*, *J Real-Time Image Proc*, doi: <https://doi.org/10.1007/s11554-016-0622-y>, p. 1–13 (Online first)
- Hosny KM, Darwish MM (2017) Invariant image watermarking using accurate polar harmonic transforms. *Comput Electr Eng* 62:429–447
- Hussein JA (2012) Luminance-based embedding approach for color image watermarking. *Int J Image Graph Signal Process* 4(3):49–56
- Ismail IA, Shouman MA, Hosny KM, Abdel-Salam HM (2010) Invariant image watermarking using accurate Zernike moments. *J Comput Sci* 6(1):52–59
- Kumar C, Kumar Singh A, Kumar P (n.d.) A recent survey on image watermarking techniques and its application in e-governance, *multimedia Tools Appl*, Springer DOI: <https://doi.org/10.1007/s11042-017-5222-8>
- Kuo-Cheng L (2010) Wavelet-based watermarking for color images through visual masking. *AEU-Int J Electron Commun* 64(2):112–124
- Lin P-Y, Lee J-S, Chang C-C (2011) Protecting the content integrity of digital imagery with fidelity preservation. *ACM Trans Multimed Comput Commun Appl*, Article 15 7(3):20
- Niu PP, Wang XY, Yang YP, Lu MY (2011) A novel color image watermarking scheme in non-sampled contourlet domain. *Expert Syst Appl* 38(3):2081–2098

19. Niu P, Wang P, Liu Y, Yang H, Wang X (2015) Invariant color image watermarking approach using quaternion radial harmonic Fourier moments, *Multimed. Tools Appl*
20. Pan JS, Hsin YC, Huang HC, Huang KC (2004) Robust image watermarking based on multiple description vector quantisation. *Electron Lett* 40(22):1409–1410
21. Peng H, Wang J, Wang WX (2010) Image watermarking method in multi-wavelet domain based on support vector machines. *J Syst Softw* 83(8):1470–1477
22. Ren WH, Li X, Lu ZM (2017) Reversible data hiding scheme based on fractal image coding. *J Inf Hiding Multimedia Signal Process* 8(3):544–550
23. Sebastiano B, Sabu E, Adrian U, Marcel W (2012) Multimedia in forensics, security, and intelligence. *IEEE Multimedia* 19(1):17–19
24. Shokrollahi Z, Yazdi M (2017) A Robust Blind Watermarking scheme based on stationary wavelet transform. *J Inf Hiding Multimedia Signal Process*, vol. 8, no. 3, pp. 676–687
25. Singh AK Improved hybrid technique for robust and imperceptible multiple watermarking using medical images. *Multimedia Tools Appl* 76(6):8881–8900
26. Singh C, Ranade SK (2013) Geometrically invariant and high capacity image watermarking scheme using accurate radial transform. *Opt Laser Technol* 54:176–184
27. Singh AK, Kumar B, Dave M, Mohan A (2014) Robust and imperceptible dual watermarking for telemedicine applications. *Wirel Pers Commun* 80(4):1415–1433
28. Singh A.K, Kumar B, Mohan A (2017) Medical image watermarking: techniques and applications”, book series on multimedia systems and applications. Springer, USA
29. Singh AK, Kumar B, Singh SK, Ghrera SP, Mohan A Multiple watermarking technique for securing online social network contents using back propagation neural network. *Futur Gener Comput Syst*:1–16. <https://doi.org/10.1016/j.future.2016.11.023>
30. Srivastav R, Kumar B, Kumar Singh A, Mohan A (n.d.) Computationally efficient joint imperceptible image watermarking and JPEG compression: a green computing approach, *Multimedia Tools and Applications*, Springer US DOI: <https://doi.org/10.1007/s11042-017-5214-8>
31. Suk T, Flusser J (2009) Affine moment invariants of color images, in: *The 13th International Conference on Computer Analysis of Images and Patterns*, Lecture Notes Computer Science, vol. 5702, Münste, Germany, pp. 334–341
32. Tsougenis ED, Papakostas GA, Koulouriotis DE, Tourassis VD (2013) Towards adaptivity of image watermarking in polar harmonic transforms domain. *Opt Laser Technol* 54:84–97
33. Tsougenis ED, Papakostas GA, Koulouriotis DE, Karakasis EG (2014) Adaptive color image watermarking by the use of quaternion image moments. *Expert Syst Appl* 41(14):6408–6418
34. Tsui TK, Zhang XP, Androutsos D (2008) Color image watermarking using multidimensional Fourier transforms. *IEEE Trans Inf Forensics Secur* 3(1):16–28
35. Wang Z, Bovik AC, Sheikh HR, Simoncelli EP (2004) Image quality assessment: from error visibility to structural similarity. *IEEE Trans Image Process* 13(4):600–612
36. Wang XY, Yang YP, Yang HY (2010) Invariant image watermarking using multi-scale Harris detector and wavelet moments. *Comput Electr Eng* 36(1):31–44
37. Wang XY, Niu PP, Yang HY, Wang CP, Wang AL (2014) A new robust color image watermarking using local quaternion exponent moments. *Inf Sci* 277:731–754
38. Wang C-p, Wang X-y, Zhi-qiu X (2016) Geometrically invariant image watermarking based on fast radial harmonic Fourier moments. *Signal Process Image Commun* 45:10–23
39. Wang XY, Yang HY, Niu PP, Wang CP (2016) Quaternion exponent moments based robust color image watermarking. *J Comput Res Dev* 53:651–665
40. Wang XY, Liu YN, Han MM, Yang HY (2016) Local quaternion PHT based robust color image watermarking algorithm. *J Vis Commun Image Represent* 38:678–694
41. Xiao B, Wang G, Li W (2014) Radial shifted Legendre moments for image analysis and invariant image recognition. *Image Vis Comput* 32(12):994–1006
42. Xin Y, Liao S, Pawlak M (2007) Circularly orthogonal moments for geometrically robust image watermarking. *Pattern Recogn* 40:3740–3752
43. Xin Y, Pawlak M, Liao S (2007) Accurate computation of Zernike moments in polar coordinates. *IEEE Trans Image Process* 16(2):581–587
44. Yang H, Zhang Y, Wang P, Wang X, Wang C (2014) A geometric correction based robust color image watermarking scheme using quaternion exponent moments. *Optik* 125:4456–4469
45. Yang HY, Wang XY, Wang P, Niu PP (2015) Geometrically resilient digital watermarking scheme based on radial harmonic Fourier moments magnitude. *AEU - Int J Electron Commun* 69:389–399
46. Yang HY, Wang XY, Niu PP, Wang AL (2015) Robust color image watermarking using geometric invariant quaternion polar harmonic transform. *ACM Trans Multimed Comput Commun Appl* 11(3):1–26

47. Yu M, Wang J, Jiang G, Peng Z, Shao F, Luo T (2015) New fragile watermarking method for stereo image authentication with localization and recovery. *AEU Int J Electron Commun* 69(1):361–370
48. Zear A, Singh AK, Kumar P (2016) A proposed secure multiple watermarking technique based on DWT, DCT and SVD for application in medicine. *Multimedia Tools Appl*. doi:<https://doi.org/10.1007/s11042-016-3862-8>
49. Zhang H, Shu H, Coatrieux G (2011) Affine Legendre moment invariants for image watermarking robust to geometric distortions. *IEEE Trans Image Process* 20(8):2189–2199
50. Zhu HQ, Liu M, Li Y (2010) The RST invariant digital image watermarking using radon transforms and complex moments. *Digital Signal Processing* 20(6):1612–1628



Khalid M. Hosny is a professor of information technology, faculty of Computers and Informatics at Zagazig University. He received the B.Sc., M.Sc. and Ph.D. from Zagazig University, Zagazig, Egypt in 1988, 1994, and 2000 respectively. From 1997 to 1999 he was a visiting scholar, University of Michigan, Ann Arbor and University of Cincinnati, Cincinnati, USA. He is a senior member of ACM and a member of IEEE and the IEEE computer society. His research interests include image processing, pattern recognition and computer vision. Dr. Hosny published more than 50 papers in international journals. He is an editor and scientific reviewer for more than 40 international journals.



Mohamed M. Darwish is assistant professor of Computer science, faculty of science, Assiut University, Assiut, Egypt. He received the B.Sc. (Hons.), M.Sc. and Ph.D. in Computer Science from Faculty of Science, Assiut University. His research area is Image Processing, Association Rule Mining and Medical Image Analysis.